MDP Project — What We Did

Our project studies air pollution across 204 Indian monitoring stations using machine learning. We built the full system, tested it honestly, and wrote it up as a research paper. Below, in simple words, are the **gaps we covered** and the **languages and tools we used**.

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1. Gaps we covered (problems we made sure to solve)

Gap	What we did about it, in simple words
Honest testing	Many studies test models the easy way, which fakes high scores. We tested the proper way (by time and by station), so our accuracy numbers are real, not inflated.
Cause vs. coincidence	We did not assume that two things moving together means one causes the other. We checked each link before calling it a cause.
Weather check	Weather drives a lot of pollution. We checked for this, so we don't wrongly give weather's effect to the pollutants.
Grouping is not sources	We were clear that grouping similar pollutants is NOT the same as finding where they come from (real source-finding needs more data).
Model vs. real policy	We showed that lowering a number inside a model is not the same as a real-world emission cut, so we don't over-claim policy advice.
Clean data & sharing	We checked the measurement units properly (and found a real unit problem) and made the whole dataset and code open and reusable.

2. Languages & tools we used (and why)

Tool / language	What it is for, in simple words
Python	The main programming language — easy to read and the standard choice for data and AI work.
PySpark (Apache Spark)	Handles very large data (over 1 million rows) quickly by splitting the work.
XGBoost	The AI model that learns to predict pollution levels.
DirectLiNGAM	Looks for cause-and-effect links between pollutants.
Optuna / NSGA-II	Searches for the best trade-off ‘policies’ between two goals.
SHAP	Explains which factors the AI model relies on most.
Matplotlib	Draws the charts and figures in the paper.
Git & GitHub	Saves every version of the code and lets the team share it.
Google Cloud (Dataproc)	Runs the whole pipeline on the cloud, not just one laptop.
LaTeX	Types the final research paper neatly, the way journals expect.

In short: we covered six common mistakes that weaken air-quality AI studies, and we used standard, widely-trusted tools — so our results are honest, reproducible, and ready to write up.